

Deokjae Lee

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SUMMARY

My research develops scalable optimization algorithms for practical machine learning problems, so far focusing on high-dimensional black-box and discrete settings, with contributions to LLM quantization, neural network compression, black-box attacks and red teaming, and biological sequence design.

EDUCATION

Seoul National University, M.S. & Ph.D. Sep.2020 - Aug.2026
Department of Computer Science and Engineering (Advisor: Prof. Hyun Oh Song)

Seoul National University, B.S. Mar.2016 - Aug.2020
Department of Mathematical Sciences

Seoul Science High School Mar.2012 - Feb.2015

EXPERIENCES

AI Institute, Seoul National University Sep.2026 - Current
Postdoctoral Researcher
Currently improving the memory efficiency of the model agent.

Center for Data Science, New York University Sep.2023 - Feb.2024
Visiting Scholar (Advisor: Prof. Kyunghyun Cho)
Conducted research on active learning algorithm for multi-objective combinatorial problem [C5].

DeepMetrics Jul.2023 - Aug.2023
Research Intern (Advisor: Prof. Hyun Oh Song)
Evaluated AI for ventilator control via causal inference models.

PUBLICATIONS

- C: Conference, P: Preprint, *: equal contribution.

[P1] Weighted Maximum Antichain Identification Algorithm through a Linear Programming
Junhyeok Lee, **Deokjae Lee**, Yeonwoo Jeong, Hyun Oh Song

[C8] Q-Strata: Hierarchical Bit Allocation for Mixed-Precision Quantization of Mixture-of-Experts LLMs
Deokjae Lee, Sihun Chu, Hyun Oh Song
EMNLP 2026.

[C7] Q-Palette: Fractional-Bit Quantizers Toward Optimal Bit Allocation for Efficient LLM Deployment
Deokjae Lee, Hyun Oh Song
NeurIPS 2025.

[C6] GuidedQuant: Large Language Model Quantization via Exploiting End Loss Guidance
Jinuk Kim, Marwa El Halabi, Wonpyo Park, Clemens JS Schaefer, **Deokjae Lee**, Yeonhong Park, Jae W. Lee, Hyun Oh Song
ICML 2025.

[C5] Training Greedy Policy for Proposal Batch Selection in Expensive Multi-Objective Combinatorial Optimization
Deokjae Lee, Hyun Oh Song, Kyunghyun Cho
ICML 2024.

[C4] Query-Efficient Black-Box Red Teaming via Bayesian Optimization

Deokjae Lee, JunYeong Lee, Jung-Woo Ha, Jin-Hwa Kim, Sang-Woo Lee, Hwaran Lee, Hyun Oh Song
ACL 2023.

[C3] Efficient Latency-Aware CNN Depth Compression via Two-Stage Dynamic Programming
Jinuk Kim*, Yeonwoo Jeong*, **Deokjae Lee**, Hyun Oh Song
ICML 2023.

[C2] Query-Efficient and Scalable Black-Box Adversarial Attacks on Discrete Sequential Data via
Bayesian Optimization
Deokjae Lee, Seungyong Moon, Junhyeok Lee, Hyun Oh Song
ICML 2022.

[C1] Optimal channel selection with discrete QCQP
Yeonwoo Jeong*, **Deokjae Lee***, Gaon An, Changyong Son, Hyun Oh Song
AISTATS 2022.

TEACHING

Teaching Assistant, Introduction to Deep Learning (M2177.0043)	Spring 2023
Teaching Assistant, Machine Learning (4190.666)	Fall 2020
Undergraduate Student Instructor, Basic Calculus 2 (033.017)	Fall 2017
Undergraduate Student Instructor, Basic Calculus 1 (033.016)	Spring 2017

HONORS AND AWARDS

Qualcomm Innovation Fellowship Korea, Finalist [C4]	2023
Yulchon AI Star Scholarship	2023
Qualcomm Innovation Fellowship Korea, Finalist [C2]	2022
Silver Medal , Korean Contest of Mathematics for University Students	2019
National Science & Technology Scholarship	2018 - 2020
Silver Medal , Korean Mathematical Olympiad (KMO)	2013

ACADEMIC SERVICES

Conference Reviewer: NeurIPS (2022-2026), ICML (2023-2026), ICLR (2024-2025), AAAI (2024,2026),
ACL (2023), EMNLP (2023), COLM (2024,2026).

Journal Reviewer: TPAMI (2024), TMLR (2026).

SKILLS

Programming Languages and Frameworks

- Advanced: Python, PyTorch, TensorFlow, LaTeX.
- Intermedicate: C++, MATLAB.

PATENTS

Method and apparatus with text classification model

- Jongseok Kim, Hyun Oh Song, **Deokjae Lee** (US Patent Application 18/336,578)

REFERENCES

Hyun Oh Song Associate Professor, Dept. of Computer Science and Engineering, SNU (hyunoh@snu.ac.kr).

Kyunghyun Cho Professor, Dept. of Computer Science and Engineering, NYU (kc119@nyu.edu).